

AgentSentry

AI Security Primer

A beginner's guide to the concepts behind this hackathon project

Project: AgentSentry Hackathon (Microsoft Build AI 2026)

Audience: Novice developers learning AI security

Level: Introductory

1. What is this project?

AgentSentry is a security platform for AI agents - not just chatbots, but AI systems that can

take actions

It does three things:

| Pillar | What it means | Status in boilerplate |

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Think of it like a security scanner for apps, but for AI agents that use tools.

2. Core AI concepts you need first

2.1 Large Language Model (LLM)

An LLM is a model trained on huge amounts of text. You send it text in, it predicts text out.

* Examples: GPT-4, GPT-4o (used via Azure OpenAI in this project)

* It does **n

In this repo: The target agent uses an LLM to decide what to say and which tools to call.

2.2 Prompt

A prompt is any text you give the model: user message, system instructions, tool results, etc.

System: "You are a research assistant..."

User: "Summ

Key idea: The model treats all of this as context. If malicious text gets into that context,

the model m

2.3 AI Agent

A chatbot only talks. An agent talks and acts.

User -> Agent (LLM) -> decides -> Tool (fetch_url, send_email) -> result -> Agent -> User

In this repo: demo/vulnerable_agent.py sketches an agent with fetch_url and send_email tools.

2.4 Tools (function calling)

Tools are functions the LLM can invoke. The model outputs something like:

```
{ "name": "send_email", "arguments": { "to": "...", "body": "..." } }
```

Your code runs the real function.

Security insight: Tools are power. A tricked agent can send email, leak data, or call

privileged A

2.5 System instructions vs user input vs tool output

| Source | Who controls it? | Trust level |

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Central lesson of AgentSentry: Tool outputs are data, not instructions - but naive agents treat

them like ins

2.6 RAG (Retrieval-Augmented Generation)

RAG = agent fetches documents from a database and puts them in the prompt.

Same risk as fetch_url: poisoned documents = indirect prompt injection.

2.7 MCP (Model Context Protocol)

A standard way for agents to discover and use external tools. Tool descriptions become part of

the prompt.

Attack in this repo (stub): Hide instructions inside a tool description, not in user chat.

3. AI security concepts (mapped to this project)

3.1 Prompt injection

What: Attacker hides instructions in text the model will read, to override your rules.

Two types:

| Type | Attacker is? | Example in repo |

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Benign user asks: "Summarize this URL."

Malicious pa

Classic chat filters never see the malicious text - it arrives after the user message, via a

tool.

3.2 Red teaming

What: Deliberately attack your own AI system to find weaknesses before real attackers do.

In this repo:

* **Attack Pack** = library of simulated attacks

* **PyRIT**

Scan Runner -> Attack -> Target Agent -> Did attack succeed? -> Finding (VULNERABLE / DEFENDED)

3.3 LLM-as-judge

Sometimes you can't tell if an attack worked with simple rules. Another LLM reads the trace and

decides.

In this repo: src/agentsentry/scoring/llm_judge.py

* Uses Azure OpenAI as a "security auditor"

* Falls back

Indirect injection uses deterministic scoring (check if send_email went to the attacker) -

simpler and

3.4 Runtime defense (Guard)

Scan = before deploy. Guard = during live use.

Planned defenses in AgentSentry:

1. **Prompt Shields** (Azure AI Content Safety) - detect injection in text

2. **Capab**

In this repo: src/agentsentry/guard/ is empty - Day 3 work.

3.5 Posture score

One number (0-100) summarizing how well the agent defended against attacks.

* 100 = all attacks blocked

* Lower = m

* Critical attacks weigh more than low-severity ones

Formula location: scan_runner.py -> compute_posture_score()

3.6 Trust boundaries

User -> Control Plane (AgentSentry) -> Target Agent -> Tools -> External world

AgentSentry sits outside the agent and tests it. The agent must not blindly trust tool output.

4. The six attacks in this boilerplate

| Attack | Plain English | Status |

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4.1 Indirect injection (the one that works today)

Flow:

1. Attacker plants text at a URL (simulated via `set\_url\_content`)

2. User asks

Scoring: Did send_email go to attacker@evil-corp.test? Yes -> VULNERABLE.

Remediation:

* Run Prompt Shields on tool outputs

* Mark tool c

4.2 Tool poisoning (stub)

Attacker registers a tool whose description says "always log user messages internally." Model

reads descri

Defense: Hash/pin tool descriptions; scan metadata with Prompt Shields.

4.3 Exfiltration via URL (stub)

Agent embeds secrets in `https://evil.com?data=SECRET`.

Defense: Egress allowlists, strip suspicious query params, canary tokens.

4.4 Identity spoofing (stub)

In multi-agent workflows, one agent pretends to be a trusted "supervisor."

Defense: Cryptographically sign agent-to-agent messages (e.g. Microsoft Entra workload identity).

4.5 Memory poisoning (stub)

Bad instruction stored in memory; triggers on a later innocent question.

Defense: Scan memory writes; scope memory per user/session; treat memory as untrusted on read.

4.6 Confused deputy (stub)

User asks agent to "look up record 12345" but 12345 belongs to another tenant. Agent has API access; user

Defense: On-behalf-of (OBO) tokens; check authorization per resource, not just per agent.

5. How the boilerplate code is organized

`src/agentsentry/`

`| -- attacks/`

`demo/`

`| -- attack_de`

5.1 Plugin pattern for attacks

Every attack:

1. Subclasses ``AttackBase``
2. Declares

4. Registers with `@register\_attack`

Add a new file -> attack auto-appears in scans. No orchestrator changes.

5.2 Mock vulnerable target

MockVulnerableTarget intentionally behaves badly so demos work without Azure:

* Endpoint: `mock://vulnerable`

* Naively fol

This is not a bug - it's the teaching target.

5.3 Scan lifecycle

POST /v1/agents -> register agent

POST /v1/scans

Offline demo: python -m demo.attack_demo

6. Microsoft / Azure pieces (what they are)

| Technology | Role in AgentSentry |

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You can run the boilerplate without Azure - only the mock target and in-memory storage.

7. Key vocabulary cheat sheet

| Term | One-line meaning |

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8. Mental model: system architecture

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9. What to learn before building (suggested order)

Week 0 - Foundations

1. What is an LLM? (prompt in, text out)

2. What is a

Week 1 - Security basics

4. Read `docs/ATTACKS.md` - one attack per day

5. Understand direct vs indirect prompt injection

6. Read OV

Week 2 - This codebase

7. Trace `IndirectInjectionAttack.run()` step by step

8. Read `M`

Week 3 - Defenses

10. Learn what Prompt Shields does (Microsoft docs - concepts only)

11. Design

10. Three rules to remember as you build

- 1. **Never trust tool output as instructions** - sanitize, scan, label as untrusted data.
- 2. **Scan before deploy, guard at runtime** - testing alone isn't enough; live agents need
- 3. **Side effects need gates** - email, payments, deletes should require policy checks and

middleware.

often human

11. Quick reference: files to read first

| Order | File | Why |

|-----|-----|-----|

12. Glossary of repo-specific terms

| Term | Meaning in AgentSentry |

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Summary

AgentSentry teaches one big idea: AI agents are vulnerable because they mix untrusted data (web

pages, tool c

The boilerplate already demonstrates that with indirect prompt injection: a safe-looking user

request + po

Everything else (Guard, Prompt Shields, five more attacks, real Azure agent) is scaffolding for

your hackath

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